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Lab Assignment #08

Aim:

Design a microstrip patch antenna at frequency 2.4 GHz, using Alumina substrate with 1mm thickness, and micro stripline characteristic impedance of 60 Ohm.

Theory:

A microstrip patch antenna is a low-profile, highly versatile radiating structure consisting of a planar radiating element (the "patch") located on one side of a dielectric substrate, with a continuous metallic ground plane on the opposite side.

Radiation Mechanism: Unlike wire antennas, patch antennas do not radiate directly from the conductive material itself. Instead, radiation occurs due to the fringing electric fields that exist at the open edges of the patch. When excited by an alternating current, the region between the patch and the ground plane acts as a resonant cavity. The electric fields "bulge" or fringe outward from the edges of the copper into the surrounding air and dielectric before terminating on the ground plane. These fringing fields act as parallel radiating slots, launching electromagnetic waves into free space.

Impedance Matching and Inset Feeding :The input impedance of a patch antenna varies significantly across its surface. It is maximum at the radiating edges (typically between 200 Ω and 600 Ω) and drops to zero at the exact geometric center. To achieve maximum power transfer from a standard feedline (e.g., 60 Ω), an inset feed is utilized. By cutting a notch into the patch, the feedline connects directly to a point inside the patch where the internal resistance perfectly matches the characteristic impedance of the feedline, thereby minimizing the Voltage Standing Wave Ratio (VSWR).

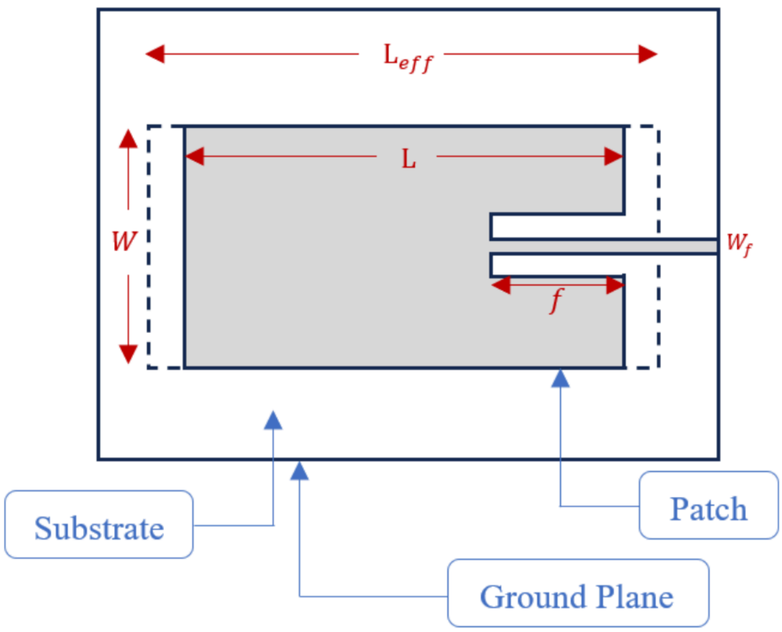

Fig 1.1 Micro Stripline Patch Diagram

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Design Parameters and Mathematical Formulation

The fundamental constants used for this design are:

- **Operating Frequency (f0):** 2.4 GHz
- **Speed of Light (c):** 3 x 10^8 m/s
- **Substrate Thickness (h):** 1.0 mm
- **Dielectric Constant (ϵr):** 9.9 (Alumina)
- **Target Impedance (Zin)**: 60 Ω
- **Feedline Width (Wf):** 0.6362 mm

Patch Width

The width dictates the radiation efficiency and the edge impedance of the antenna.

$$W = \frac{c}{2f_0}\sqrt{\frac{2}{\epsilon_r + 1}} = \frac{300}{2 \times 2.4}\sqrt{\frac{2}{9.9 + 1}} = 26.75 \text{ mm}$$

Effective Dielectric Constant

Because the fringing fields travel partially in the dielectric and partially in the air, the effective permittivity is lower than the substrate's bulk permittivity.

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}\left[1 + 12\frac{h}{W}\right]^{-0.5} = \frac{9.9 + 1}{2} + \frac{9.9 - 1}{2}\left[1 + 12\left(\frac{1}{26.75}\right)\right]^{-0.5} = 9.15$$

Fringing Length Extension (ΔL)

The fringing fields make the antenna appear electrically longer than its physical dimensions.

$$\Delta L = 0.412h\frac{(\epsilon_{reff} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\epsilon_{reff} - 0.258)\left(\frac{W}{h} + 0.8\right)} = 0.412(1)\frac{(9.15 + 0.3)\left(\frac{26.75}{1} + 0.264\right)}{(9.15 - 0.258)\left(\frac{26.75}{1} + 0.8\right)} = 0.43 \text{ mm}$$

Theoretical Patch Length (L)

The length establishes the primary half-wavelength resonance. The fringing extensions on both sides must be subtracted from the total effective length.

$$L = \frac{c}{2f_0\sqrt{\epsilon_{reff}}} - 2\Delta L = \frac{300}{2(2.4)\sqrt{9.15}} - 2(0.43) = 19.79 \text{ mm}$$

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Radiating Edge Impedance

Using the tuned dimensions from the 3D model (L=19.79 mm, W=26.75 mm), the radiation resistance at the edge of the patch is calculated to establish the top of the impedance curve.

$$R_{edge} = 90\left(\frac{\epsilon_r^2}{\epsilon_r - 1}\right)\left(\frac{L}{W}\right)^2 = 90\left(\frac{9.9^2}{9.9 - 1}\right)\left(\frac{19.79}{26.75}\right)^2 = 542\ \Omega$$

Inset Feed Length

The physical distance from the edge of the patch to the 60 Ω match point, following the cosine-squared impedance distribution.

$$y_0 = \frac{L}{\pi}\cos^{-1}\left(\sqrt{\frac{Z_{in}}{R_{edge}}}\right) = \frac{19.79}{\pi}\cos^{-1}\left(\sqrt{\frac{60}{542}}\right) = 7.759\ \mathrm{mm}$$

Total Inset Width

To cancel parasitic inductance created by the cutout, a capacitive gap equal to half the feedline width is placed on both sides of the feedline.

$$\text{Inset Width} = 2 \times W_f = 2 \times 0.6362 = 1.2724\ \mathrm{mm}$$

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Final Simulation Parameter Table

The following table summarizes the final dimensions applied to the 3D electromagnetic simulation model to achieve precise resonance and impedance matching at 2.4 GHz.

Table 1 - Parametric Details of 3D Model

Parameter Description	Final Value
Operating Frequency	2.4 GHz
Dielectric Constant (Alumina)	9.9
Target Input Impedance	60 ohm
Substrate Width	50.0 mm
Substrate Length	50.0 mm
Substrate Thickness	1.0 mm
Patch Width	26.75 mm
Patch Length	19.79 mm
Feedline Width	0.6362 mm
Feedline Length	15.105 mm
Inset Feed Length	7.759 mm
Total Inset Cutout Width	1.2724 mm
Copper Patch Thickness	0.035 mm
Copper Ground Thickness	0.035 mm

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Procedure:

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Navigate to the Microwave & RF / Optical folder, choose Antennas, and then select Planar (Patch / Slot).
3. Choose the Time Domain Solver from the available options to handle the broadband frequency sweep.
4. Set the project units by defining Dimensions in mm and Frequency in GHz.
5. Set the simulation range with a Minimum of 1 GHz and a Maximum of 5 GHz to fully observe the 2.4 GHz resonance and impedance bandwidth.
6. Open the Modeling tab and set the Background Material to Normal (Air/Vacuum) with open (add space) boundary conditions in all directions to allow for free-space radiation.
7. Create parameters in the Parameter List for the substrate dimensions, patch dimensions (patch_width=26.75, patch_length=19.79), feedline dimensions, and inset cuts (inset_length=7.759, inset_width=1.2724).
8. Use the Brick tool to create the main substrate layer with dimensions based on substrate_width and substrate_length, assigning the material as Alumina (loss free).
9. Create a second Brick for the ground plane on the bottom face of the substrate, assigning the material as Copper (Annealed) or PEC, with a thickness of 0.035 mm.
10. Use the Brick tool on the top face of the substrate to construct the main radiating patch and the microstrip feedline, assigning both as Copper, and merge them using the Boolean Add operation.
11. Use the Brick tool to define the inset gaps on either side of the feedline, and use the Boolean Subtract operation to carve these capacitive notches out of the main copper patch.
12. Use the Pick Face tool to select the front cross-sectional face of the feedline, and define a Waveguide Port sized appropriately to capture the fringing fields (e.g., expanding the port width and height beyond the copper trace).
13. Navigate to the Simulation tab and define Field Monitors for the Electric (E) field, Magnetic (H) field, and Farfield (radiation pattern) specifically at 2.4 GHz.
14. Open the Setup Solver settings, change the Accuracy from the default to -40 dB or -50 dB to ensure the highly resonant energy fully dissipates, and start the simulation.
15. Navigate to the 1D Results folder to view S-parameter (S11) and VSWR plots, and the 2D/3D Results folder to visualize the directivity and field distributions at the monitored 2.4 GHz frequency.

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<u>3D Model:</u>

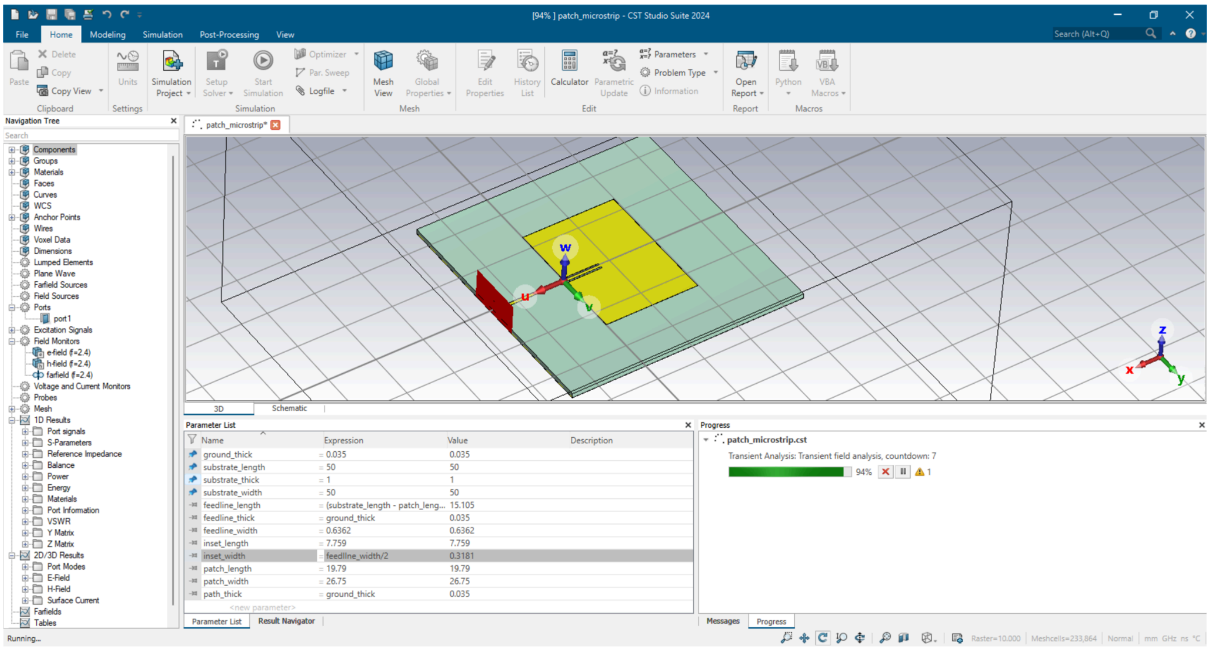

Fig 2.1 Micro Stripline Patch 3D Model with parameters

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Results:

Scattering Parameters

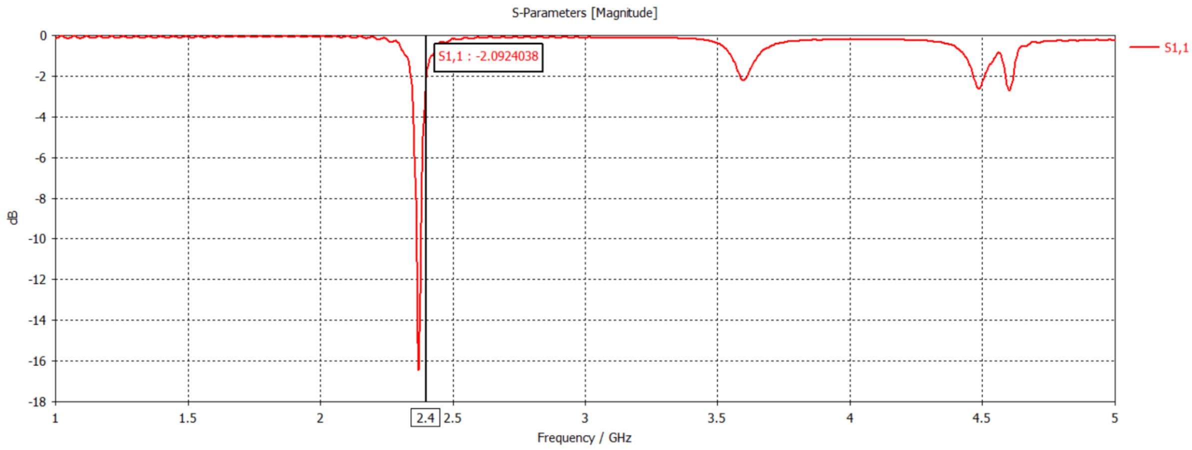

Fig 3.1 S parameter of micro stripline patch at 2.4 GHz frequency

Reference Impedance

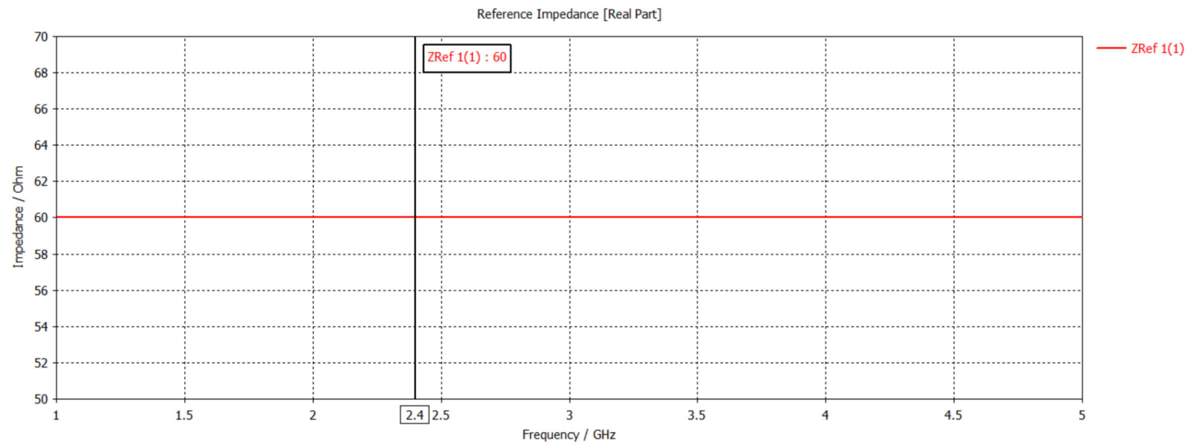

Fig 3.2 Observed reference impedance

Voltage Standing Wave Ratio (VSWR)

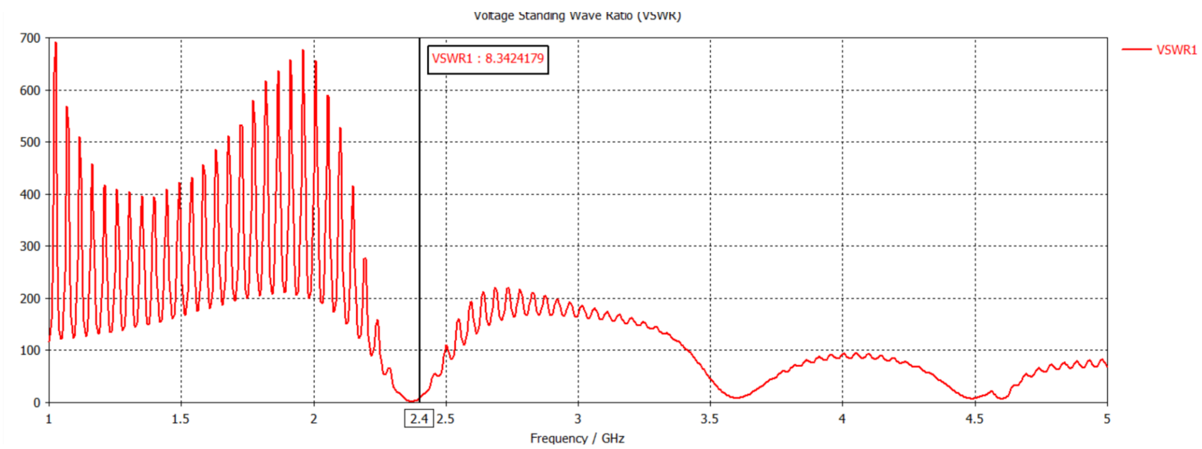

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Fig 3.3 VSWR of micro stripline patch at 2.4 GHz frequency

Farfields

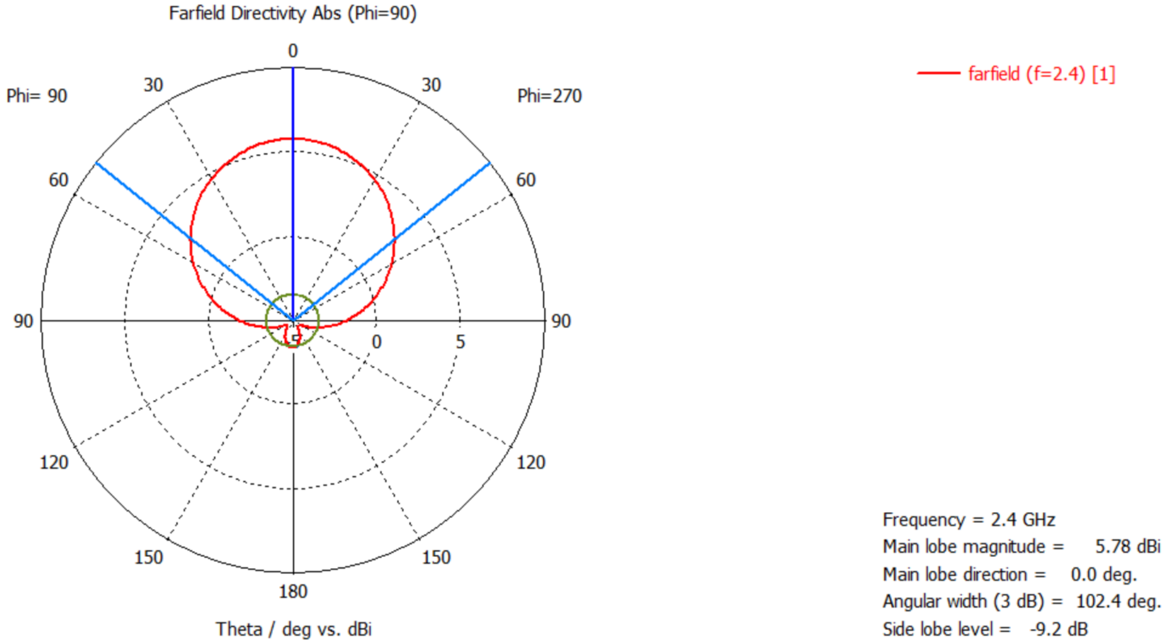

Fig 3.4 Farfield distribution at 2.4 GHz frequency

Electric Field Distribution

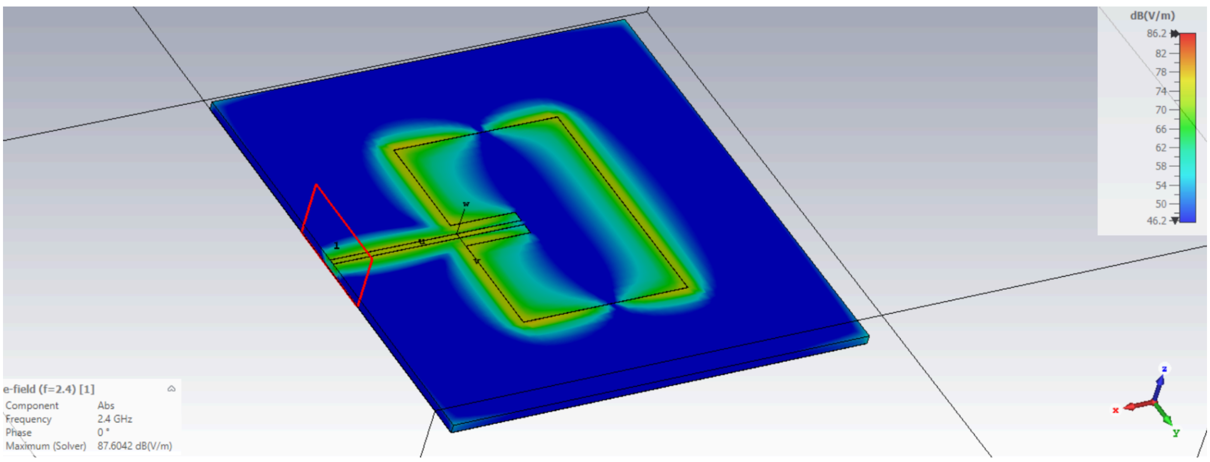

Fig 3.5 Electric field distribution at 2.4 GHz frequency

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Magnetic Field Distribution

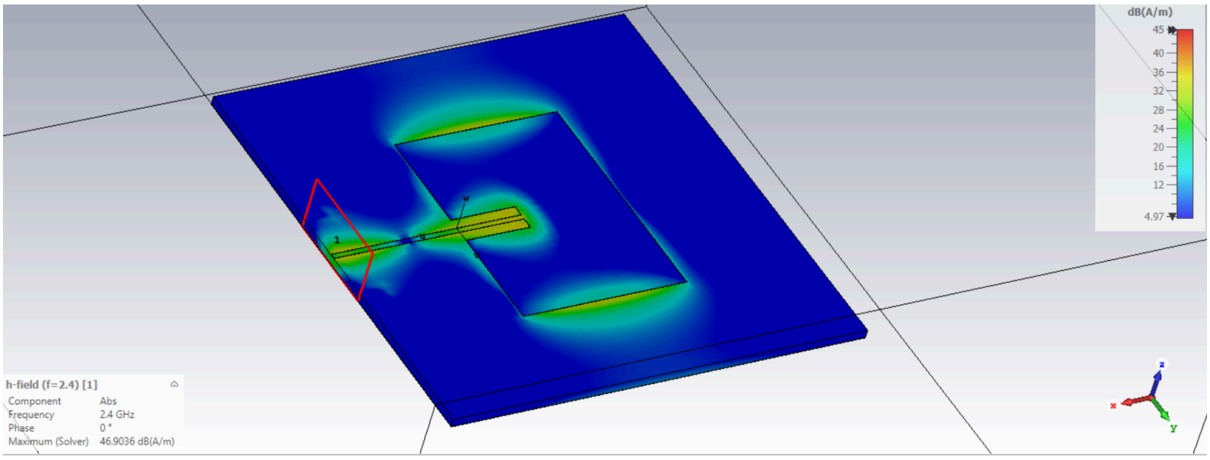

Fig 3.6 Magnetic field distribution at 2.4 GHz frequency

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Conclusions:

In this experiment, a rectangular microstrip patch antenna operating at 2.4 GHz was successfully designed, mathematically modeled, and simulated using CST Studio Suite. The antenna was modeled on a 1.0 mm thick Alumina substrate ($\epsilon r=9.9$) and fed using a 60 Ω microstrip transmission line.

The results demonstrate that while standard analytical formulas—such as the transmission line model for calculating width, effective dielectric constant, and fringing extensions—provide a robust foundation for the initial geometry, 3D full-wave simulation is strictly required for final optimization. The dense Alumina substrate tightly bound the electromagnetic fields, making the design highly sensitive to small dimensional changes. Through parametric tuning of the patch length (L), the resonant frequency was precisely shifted to 2.4 GHz, compensating for the complex 3D fringing effects not fully captured by theoretical equations.

Furthermore, the experiment validated the effectiveness of the inset feeding technique. By carefully calculating the physical penetration depth and properly sizing the capacitive gap around the feedline, the high radiation resistance at the edge of the patch (>500 Ω) was successfully stepped down to match the 60 Ω source. The final simulated VSWR and return loss confirm minimal signal reflection and efficient broadside radiation, fulfilling all the fundamental objectives of the design assignment.